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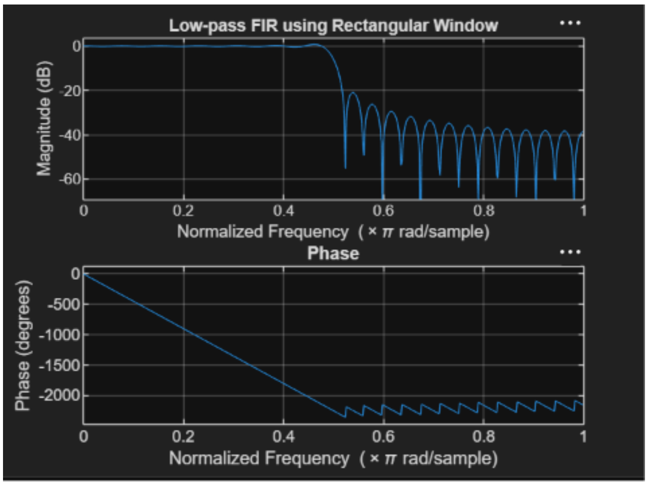
COURSE CODE: ECA0905
COURSE NAME: DIGITAL SIGNAL PROCESSING
PRACTICAL LAB RESULTS

EXP: 4.1:
CODE:

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
```

```
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

OUTPUT:




```
--- Rectangular Window Coefficients ---
    0.0126
         0
   -0.0137
         0
    0.0150
         0
   -0.0166
         0
    0.0185
         0
   -0.0210
         0
    0.0242
         0
   -0.0286
         0
    0.0349
         0
   -0.0449
         0
```

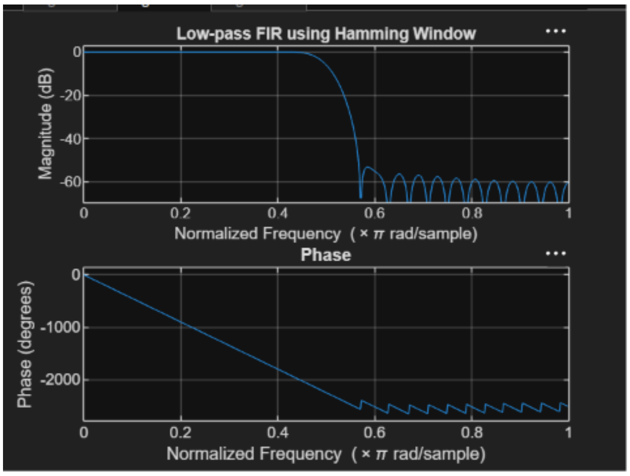
EXP:4.2:
CODE:

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)


wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');

disp(b_rect');

%% --- 2. FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm');

%% --- 3. FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann');

%% --- Plot Frequency Responses ---

figure;

freqz(b_rect,1);

title('Low-pass FIR using Rectangular Window');

figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');

figure;
```

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');

OUTPUT:




```
--- Hamming Window Coefficients ---
    0.0010
         0
   -0.0013
         0
    0.0021
         0
   -0.0034
         0
    0.0055
         0
   -0.0084
         0
    0.0125
         0
   -0.0181
         0
    0.0260
         0
   -0.0379
```

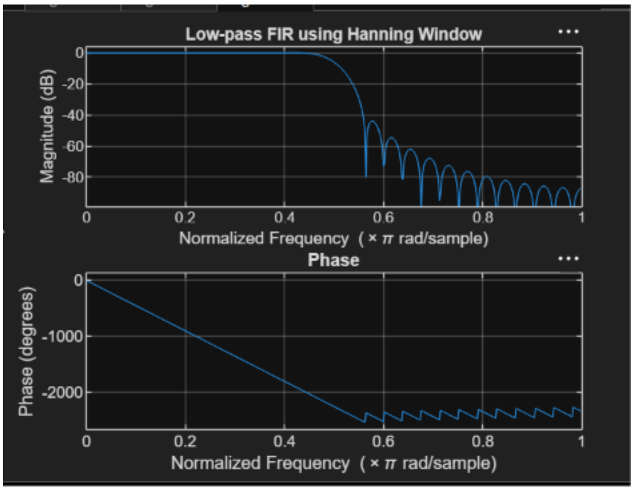
EXP:4.3:

CODE:

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
```

```
disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

OUTPUT:




```
--- Hanning Window Coefficients ---
    0.0000
         0
   -0.0004
         0
    0.0013
         0
   -0.0028
         0
    0.0050
         0
   -0.0081
         0
    0.0122
         0
   -0.0179
         0
    0.0259
         0
   -0.0378
```